Percentages and ratios — Equivalents · Percentage of amounts · Interest · Ratio

A calculator is allowed for questions marked CALC. ★ = challenge question.

A · PERCENTAGES

1 Copy and complete the table of equivalents.

Fraction		$\frac{7}{20}$			$2\frac{2}{5}$
Decimal	0.6			1.25	
Percentage			15%		

2 Three groups of students tried a new snack. Group A: 80 of 200 liked it. Group B: 33 of 60. Group C: 63 of 140.

a Write each group's proportion who liked it as a percentage.

b Which group liked it most? Which group liked it least?

3 A 750 ml fruit drink contains 300 ml of apple juice, 150 ml of mango juice and the rest is grape juice. What percentage of the drink is **a** apple, **b** mango, **c** grape?

4 A phone costs €480. In a sale it is reduced by 35%. Work out the sale price.

5 Omar gets a pay rise of 6%. His old salary was €25 000. What is his new salary?

6 **a** 10% of an amount is 4.8 kg. Work out the amount.

b 30% of an amount is 72 m. Work out the amount.

7 CALC Lena invests €2400 for 3 years at 4.5% **simple** interest per year. How much is the investment worth after 3 years?

8 CALC Kostas puts €3000 in an account paying 2% **compound** interest per year. How much is in the account after 3 years?

B · RATIOS

9 Simplify each ratio.

a $14 : 21$ **b** $2.5 : 7.5$ **c** $0.8 : 2$

10 Sophie mixes squash and water in the ratio $3 : 7$. She makes 450 ml of drink. How much squash and how much water does she use?

11 Green paint is made from blue, yellow and white in the ratio $5 : 3 : 2$. How much of each colour is needed for 2 litres of green paint?

12 An alloy is 60% copper, 30% zinc and 10% nickel.

a Write the fraction of the alloy that is each metal, in simplest form.

b Write the ratio copper : zinc : nickel in its simplest form.

13 A shop has 20% off everything. For one day it offers an extra 30% off the sale price.

a A jacket originally costs €150. Work out its price on that day.

b Is "20% off then 30% off" the same as 50% off? Explain.

14 After a 15% reduction, a pair of trainers costs €51. What was the original price?

Check: percentage = part ÷ whole × 100 · multipliers: +6% → 1.06, −35% → 0.65 · simple interest is the same every year, compound grows · share in a ratio: find one part first.